

Algebra and discrete mathematics

Student investigations / Remaining atlas entries / September 2026

Choose an investigation whose tools you know. Make things, try ideas and explain what you notice. Use scrap paper or work with a partner. Later pages can be another meeting.

AD-01	Build a world that defeats a rule	4
	Every/some language; witness objects and a complete four-type argument	
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	Addition on a four-ring, independent representatives, and pair translation	

Print selected pages, single-sided, on US Letter at actual size. Keep student pages and solutions separate. The original random-ten book remains a separate volume.

Contents continued

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The listed tools are prerequisite gates, not age labels. More detailed gates appear in the facilitator guide.

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Make the rule survive

Build a world that defeats a rule

Core: spoken every/some rules

Every card is red or blue, and a circle or square. Duplicates are allowed. An every-rule fails only when a card breaks it.

1. Build a world with a red object. Keep “Every red object is a circle” true and make “Every circle is red” false. Use the fewest cards possible. Mark the rule-breaker.

R red
circle

R red
square

B blue
circle

B blue
square

My smallest world

A different attempt

2. Choose an extra card that keeps the first rule true and one that breaks it. Describe every allowed card type.

R red
circle

R red
square

B blue
circle

B blue
square

keeps / breaks

keeps / breaks

keeps / breaks

keeps / breaks

A missing witness

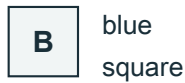
Build a world that defeats a rule

Core continuation: negation and empty classes

No red object is promised on this page. An empty tray is allowed as a game world.

5. Test “Every red object is a circle” and “There is a red circle” on one blue square and on the empty tray. Explain the difference.

One blue square



Empty tray

6. Compare “Not every red object is a circle” with “Every red object is a square.” Build a nonempty world where just the first is true. Can you make just the second true?

Only the first statement is true

Only the second statement is true

Escape a collection

Can your list catch my pattern?

Core: matching positions

Every pattern has four places. One mismatch is enough to make two patterns different. The supplied collection is 0011, 0101, 1010, 1111.

1. Choose a missing pattern. Mark a place where it differs from each row. Could another answer also win?

	1	2	3	4
1	0	0	1	1
2	0	1	0	1
3	1	0	1	0
4	1	1	1	1

My missing pattern

--	--	--	--

One witness place for each row

row	1	2	3	4
place				

2. Write four patterns for a partner; repeats are allowed. Trade lists and invent a method that always escapes without searching through every possible word.

	1	2	3	4
1				
2				
3				
4				

My answer to a partner

--	--	--	--

My guaranteed method:

Make a guarantee

Can your list catch my pattern?

Core: one witness for each row

After trying your own idea, use this method: match row 1 to position 1, row 2 to position 2, and so on. Choose the opposite bit in each matched place.

3. Apply the method to the supplied collection. Why can changes outside the matched input places not spoil it? What if you rearrange the rows and run the method again?

	1	2	3	4
1	0	0	1	1
2	0	1	0	1
3	1	0	1	0
4	1	1	1	1

Opposite bits

--	--	--	--

4. Must a list of five four-place patterns still miss something? What is the fewest rows that could catch every pattern? Build a systematic complete collection.

A complete collection, organized my way

Organize your words here; continue on scrap paper if needed

A list with no final row

Can your list catch my pattern?

Advanced: infinite sequences and arbitrary indices

Each infinite binary pattern gives a bit at every positive-integer position. A proposed complete list has rows $s_1, s_2, \dots$, each infinite. A rule at every position defines a sequence without completing an endless drawing.

5. Define a new sequence b from the list. If someone says b is row 137, where can you find a mismatch? Explain why no row number works.

A proposed infinite list

row / place	1	2	3	...
1	$s_1(1)$	$s_1(2)$	$s_1(3)$	...
2	$s_2(1)$	$s_2(2)$	$s_2(3)$	...
3	$s_3(1)$	$s_3(2)$	$s_3(3)$	...
...	...	...	...	...

My rule for position i :

6. The list maker puts b at the top and moves all old rows down. Is the collection now complete? What does your earlier proof actually promise?

New list: new witnesses

Make a large collection

A collection with no card inside another

Core: set containment

The supplied deck has every subset of A,B,C,D once, including empty. A card is inside another when all its symbols occur on the other. A legal collection never contains two distinct selected cards when one is inside the other. Sharing only some symbols is allowed.

1. Build a large legal collection. Try to improve it by trading cards, and record your best result. Explain why every pair is legal.

$\emptyset$	A	B	A B
C	A C	B C	A B C
D	A D	B D	A B D
C D	A C D	B C D	A B C D

2. What happens if you keep the empty card or ABCD? Build a legal collection with different card sizes.

A mixed-size collection

Why my selected pairs are legal

Convince a skeptic

A collection with no card inside another

Core: complete upper-bound certificate

A chain is a row where each card is inside the next. Divide the whole deck into chains, with no cards omitted or repeated. Blank chain strips are supplied; single-card chains are allowed.

3. Why can a legal collection take at most one card from a chain? Build a chain partition whose bound matches your best collection.

1

2

3

4

5

6

4. Audit a partner's certificate for nesting and full coverage. Explain why it rules out every seven-card collection, not just the ones you tried.

Keep this card

A collection with no card inside another

Core continuation: transfer the bound

The card A must now be chosen. All other rules stay the same.

5. Find the largest legal collection that keeps A. Which cards are unavailable, and which remain candidates?

A

Keep this card

My new collection

6. Prove your new answer is best using chains. Why does prescribing a three-symbol card instead of a singleton give the same maximum?

Build a chain certificate for the remaining cards

Make a message

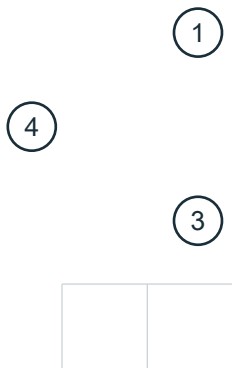
Send a network in a tiny message

Core: leaves and changing labels

A tree connects all named vertices without a cycle. A leaf has one remaining edge. Remove the smallest leaf and its edge; record its neighbor. Repeat until two vertices remain. Record the removed labels. Only numbered circles are vertices; a crossing without a circle is not a vertex.

1. Invent and encode a tree on 1,2,3,4. Then encode X with edges 12,13,34 and Y with edges 13,23,24. Show the deletion order.

My tree

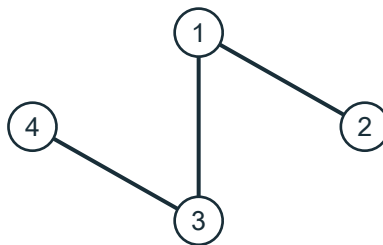


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message

removed: ___ then ___

Tree X

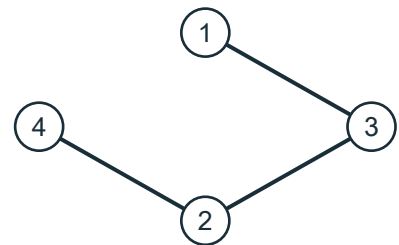


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message

removed: ___ then ___

Tree Y



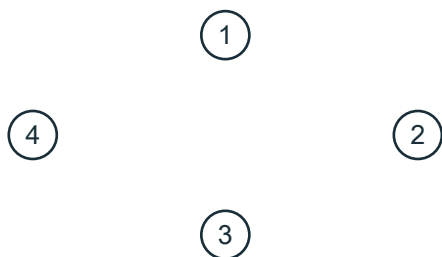
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message

removed: ___ then ___

2. Give only your message to a partner. Try to reconstruct one another's trees. What would identify the first removed leaf? Test your idea on (3,1).

Reconstruct (3, 1)



Surviving labels / next leaf / next edge

Why the receiver knows

Send a network in a tiny message

Core proof: occurrences and degree

Decoder: choose the smallest surviving label absent from the remaining message. Join it to the first entry. Remove that label and entry. Repeat, then join the final two labels. Sender and receiver both use smallest-leaf order.

3. Choose any two-entry message, perhaps a repeated one. Decode and re-encode it. Explain why every message works and why two different trees cannot send the same message.

My chosen message

surviving labels	remaining message

Decoded tree

①

④

②

③

4. Count the trees described by two-entry messages on 1,2,3,4. What do repeated entries tell you? Why is counting drawings or rotations unreliable?

A repeated-label message

①

④

②

③

A different-label message

①

④

②

③

Predict before drawing

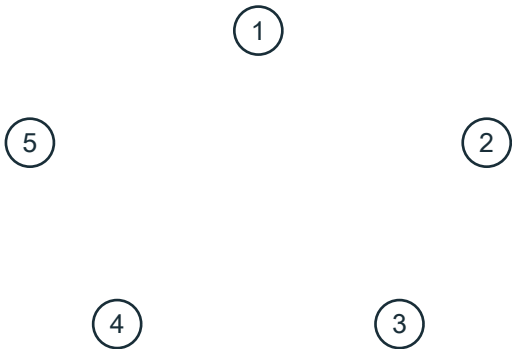
Send a network in a tiny message

Continuation: five labels; general proof is advanced

For five labels use three-entry messages. Degree is the number of edges touching a vertex. The same rules apply.

5. Decode (4,4,2). Before drawing, predict its leaves and all vertex degrees from the message. Explain the rule.

Decode (4, 4, 2)



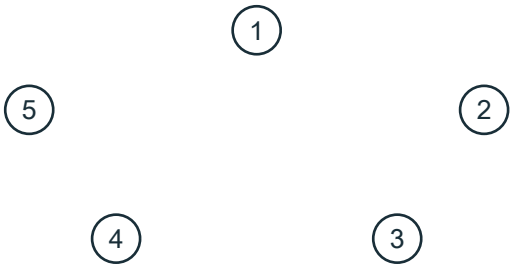
vertex	occurrences	degree
1		
2		
3		
4		
5		

6. Design a five-label message with exactly two leaves. Then, if you can justify the inverse for every $n \geq 2$, give the general count.

A message with exactly two leaves

--	--	--

Its tree



Choose the roads

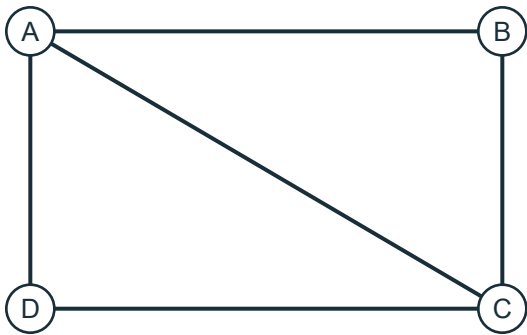
How many connecting networks survive?

Core: connectedness and unnecessary edges

Available undirected roads are AB, BC, CD, DA and AC. The four vertices are the only places. Two drawings are the same network when they keep the same named roads. A minimal connected network joins all four vertices, and removing any one kept road disconnects it.

1. Choose a minimal connected network. Try to make a different one. How many roads does each use, and why?

Available roads



Trace your chosen roads



2. Find every minimal connected network. Choose an organizing rule that helps you avoid both duplicates and omissions. Record roads, not just pictures.



Close one road

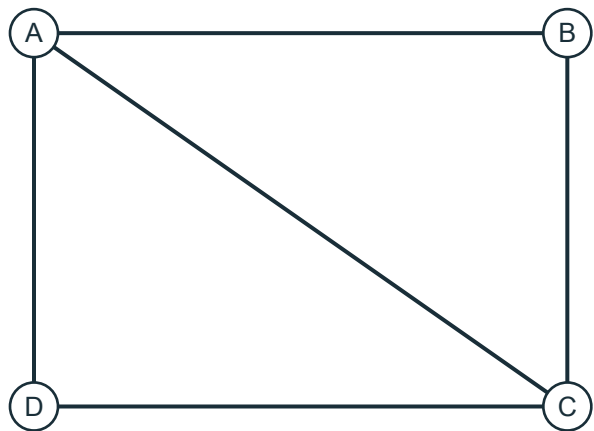
How many connecting networks survive?

Core: compare classified cases

A road closure removes that road from the available graph. A previously recorded network survives exactly when it did not use that road.

3. Choose a road to close so that as many of your networks as possible survive. Predict, then count survivors. Do all five closures have the same effect?

Choose one road to close



closed road	survivors
AB	
BC	
CD	
DA	
AC	

4. Explain the difference without recounting all eight pictures. For closure of AB, organize the surviving trees by whether they use AC.

AB closed / AC not used

AB closed / AC used

Add a road

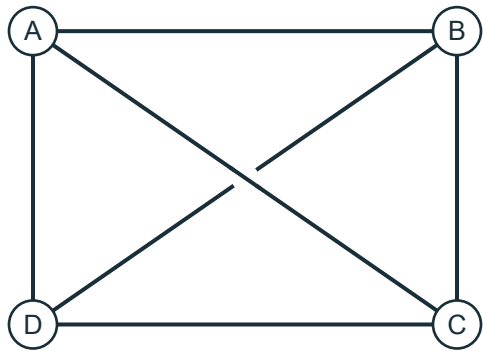
How many connecting networks survive?

Core continuation: a larger case

Now add BD. Its crossing with AC is not a new vertex: roads pass over one another. All six pairs of the four vertices are available.

5. Count the new minimal networks by separating those with no diagonal, exactly one diagonal, and both diagonals. Check that none is counted twice.

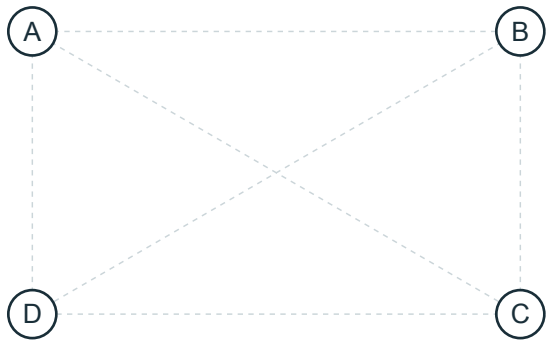
Six roads; no crossing vertex



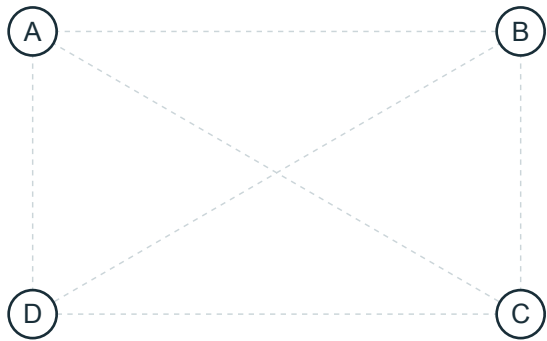
diagonals used	number of trees
none	
exactly one	
both	

6. In the six-road graph, does it matter which single road closes? Give a reason and the number of surviving trees.

One closure



Relabel it



A second counting language

How many connecting networks survive?

Advanced: matrices and determinants

Supplied theorem: for a finite connected simple graph (no loops or parallel edges), form L with vertex degrees on the diagonal and -1 for adjacent distinct vertices (0 otherwise). Delete the same row and column. The determinant equals the number of spanning trees. This page applies the theorem; it does not prove it.

7. For the original five-road graph, build L in order A,B,C,D. Delete D's row and column and compute its determinant. Compare with your classification.

vertex	degree
A	
B	
C	
D	

L

Delete D

8. Delete A's row and column instead. Why should the new determinant agree? What happens if you take $\det(L)$ without deleting anything?

Delete A

Determinant work and comparison:

Two recipes with sets

Do two calculation routes agree?

Core: union and intersection

For sets, JOIN collects every symbol in either set; MEET keeps only symbols in both. Supplied cards show all subsets of A,B,C. Recipe L: $\text{meet}(X, \text{join}(Y, Z))$. Recipe R: $\text{join}(\text{meet}(X, Y), \text{meet}(X, Z))$.

1. Choose three set cards and carry out both recipes, keeping each intermediate result visible. Try to make the answers different. Then test $X=AB$, $Y=AC$, $Z=BC$.

$\emptyset$	A	B	A B
C	A C	B C	A B C

Recipe L

JOIN(Y, Z)



MEET with X

Recipe R

MEET(X, Y)

MEET(X, Z)

JOIN the two results

2. Explain why any symbol belongs to the left answer exactly when it belongs to the right answer. Your explanation should cover every choice of three cards.

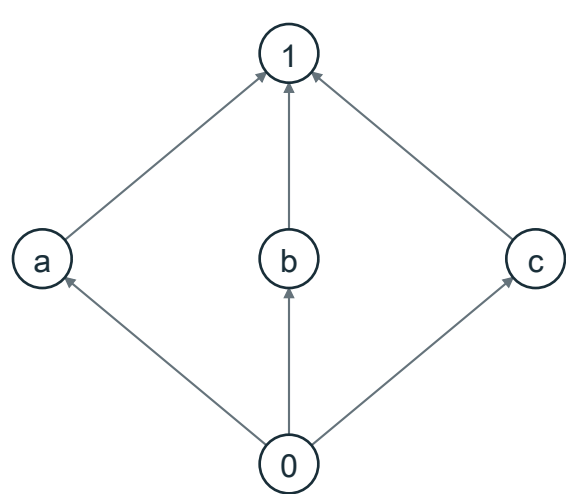
Try a new diagram

Do two calculation routes agree?

Core: greatest floor and least ceiling

The diagram has 0 below a,b,c, and 1 above a,b,c; no comparisons connect different middle tiles. Travel only upward to compare. Every tile is also above and below itself. JOIN is the least tile above both inputs; MEET is the greatest tile below both.

3. Choose several pairs and find their JOIN and MEET. Include equal inputs, two different middle tiles, and pairs involving 0 or 1. Does every pair have both operations?



pair	JOIN	MEET
same tile		
two middle tiles		
0 and a tile		
1 and a tile		

4. Choose X,Y,Z from the five tiles and try to make recipes L and R disagree. Record every intermediate result and explain whether the failure means an operation is missing.

Recipe L

JOIN(Y, Z)

↓

MEET with X

Recipe R

MEET(X, Y)

MEET(X, Z)

JOIN the two results

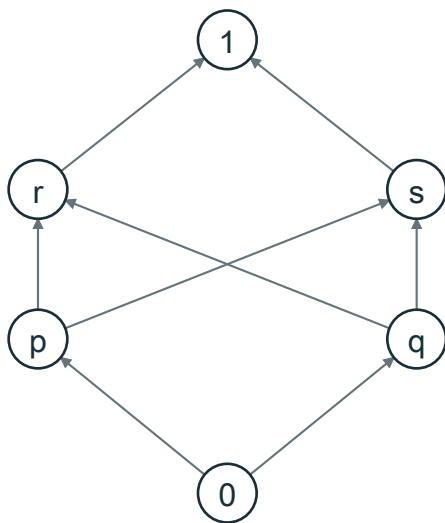
Does the operation exist?

Do two calculation routes agree?

Core continuation: least versus minimal bounds

New diagram: 0 lies below p,q; each of p,q lies below each of r,s; r,s lie below 1. The pairs p,q and r,s are incomparable. Only upward paths count.

5. Find all common upper bounds of p and q. Is one of them the least? Find the matching difficulty for $\text{MEET}(r,s)$.



Common upper bounds of p and q:

Common lower bounds of r and s:

6. Add one comparison between r and s, in either direction. Redraw or move tiles so all comparison arrows point upward. Does this repair both missing operations? Check every kind of pair to decide whether the new diagram is a lattice (every pair has both JOIN and MEET).

My added comparison and proof:

Redraw the upward diagram

Choose a starting collection

Everything your starting clues force

Core: positive rules

Rules: a gives b; b together with c gives d; d gives c. Adding does not use up prerequisites. Never remove a token or add a duplicate. Stop only when no new token can be added. Your finished set is the closure of your start.

1. Choose a starting set and finish it. Then both partners start from {a,d}, choosing different first additions. Record both runs. Do they finish differently?

a → b

b + c → d

d → c

start	first new token	next new token	finished set
{a, d}			
{a, d}			

2. Find the smallest starting collections that eventually produce all four tokens. Find every smallest one, and explain why no smaller start works.

Smallest starting collections and a reason the list is complete

Combine finished collections

Everything your starting clues force

Core: closed sets and containment

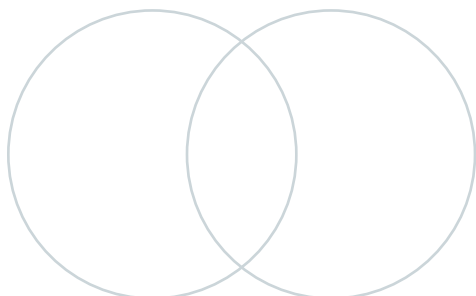
A collection is closed when no new token is forced. Union keeps every token in either collection; intersection keeps only tokens present in both.

3. Find two closed collections whose union is not closed. Show the new forced token. Then find all closed collections, using the subset cards if useful.

$\emptyset$	a	b	a b
c	a c	b c	a b c
d	a d	b d	a b d
c d	a c d	b c d	a b c d

Closed collection 1	Closed collection 2	Their union

4. Try to make the intersection of two closed collections unfinished. If it cannot happen, explain why. What is the smallest closed collection containing both {a,b} and {c}?



intersection: shared tokens

Why a rule cannot break closure:

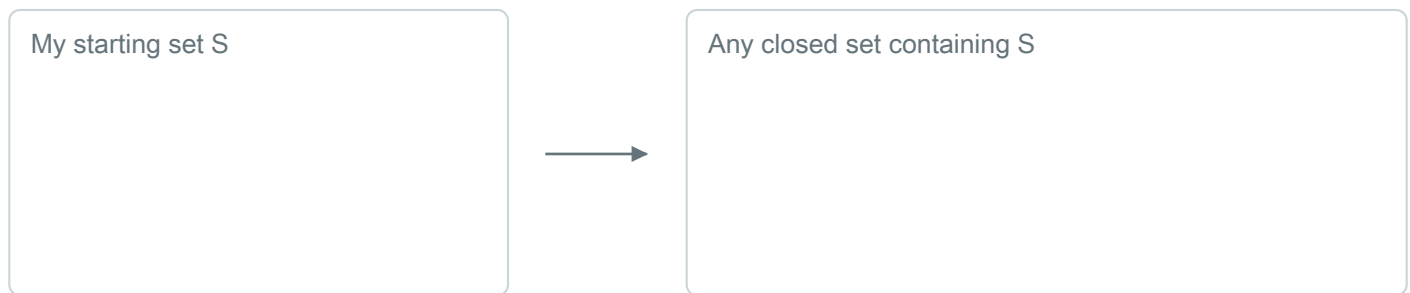
Why the order does not matter

Everything your starting clues force

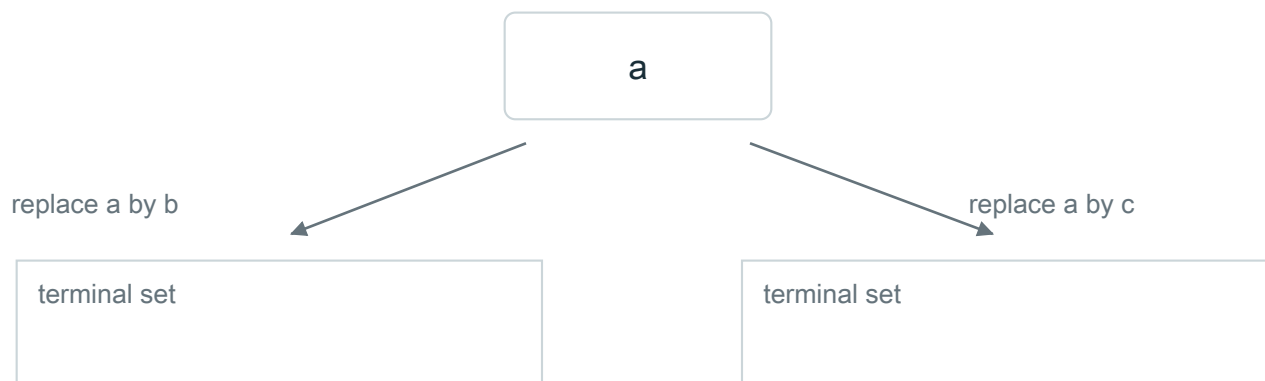
Core proof: all completed runs

A run may choose any rule that adds a new token. It is complete only when no rule can add anything. The rules remain the same throughout a run.

5. Prove that any completed run ends in the same set. Use two facts to organize your explanation: a run cannot keep adding forever, and every forced addition must belong to every closed set containing the start.



6. Change the rules: from $\{a\}$, you may replace a by b or replace a by c , removing a . Follow each first choice until no rule is possible. Explain exactly which part of the earlier guarantee fails.



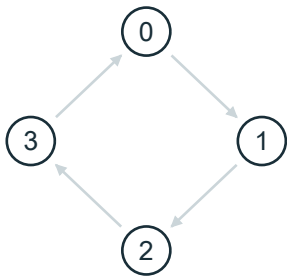
Rename the numbers

Which numbers may share a name?

Core: independent choices of representatives

Addition wraps on the ring: for example $3+2=1$. Each number belongs to one nonempty bag. To add two bags, choose a number from each, allowing the same number twice. A renaming works only if all choices give the same answer bag.

1. Invent a split into two bags and test it. Try to produce a decisive failure witness: the same input bags but two different answer bags. Test $\{0,1\}$ and $\{2,3\}$.



clockwise: add 1

+	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

My two bags

2. Find a two-bag renaming that survives every choice. Make its bag-addition table and explain why you did not need to inspect infinitely many sums.

+	bag ____	bag ____
bag ____		
bag ____		

Repair by merging

Which numbers may share a name?

Core: translation of an identified pair

You may repair a failed renaming by merging bags, but may not split bags. Use the fewest merges needed to make addition well-defined.

3. Start with $\{0,2\}\{1\}\{3\}$. What additional merge is forced? Give a same-input witness before repair and prove the repair works.

0 2

1

3

Add the same number to both: $_\sim_\$ becomes $_\sim_\$

4. Start with $\{0,1\}\{2\}\{3\}$. Determine the smallest possible repair. Explain why your argument also applies whenever neighboring ring positions share a bag.

0 1

2

3

Forced same-bag steps:

All possible renamings

Which numbers may share a name?

Core proof: complete finite classification

Every bag assignment is a partition: bags do not overlap, and all numbers are present. A valid renaming must survive addition of any fixed number to equivalent representatives.

5. Classify every valid renaming, allowing one, two, three or four bags. Give a reason that no missing case could work.

All valid partitions, with a reason there are no others

6. Add a second machine f with outputs $f(0)=0, f(1)=0, f(2)=1, f(3)=1$. A renaming must now make both addition and f independent of representatives. Which of your valid renamings survive?

input	$f(\text{input})$
0	0
1	0
2	1
3	1

Same input bag, outputs to compare:



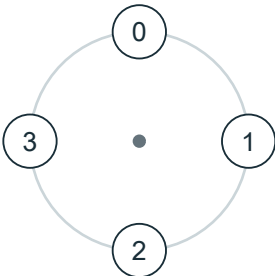
Invent a report

Can both schedules be right?

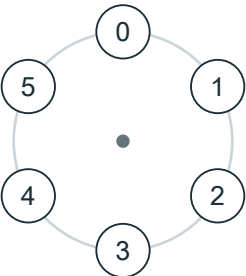
Core: remainders and synchronized movement

Clocks have positions 0,1,2,3 and 0,1,2,3,4,5. Both start at 0 and advance one step at every integer time $t \geq 0$. A report (a,b) names the four-clock first and six-clock second.

1. Choose reports that can occur together and reports that cannot. Explain each decision. Then solve the supplied report (3,5) and describe every time when it occurs.



4 positions



6 positions

Reports / times / evidence

2. A notice says (2,5). You may change exactly one reported number by one clock step, forward or backward, wrapping at zero. Find every repair, and the first time for each. Is the repaired notice uniquely determined?

Changed report	First time	Why it works

Explain all reports

Can both schedules be right?

Core proof: a complete period

A first joint return means the first positive time when both counters are back at zero. Organize the possible reports in any way that makes omissions and repetitions visible.

3. Find the first joint return. Classify every report that appears before it, without checking each pair by a separate search. Why can no report repeat earlier?

b

012345

a

0

1

2

3

Classification and non-repetition argument

4. Replace the clocks by eight and twelve positions. Predict the exact compatibility test, then test reports (5,9) and (2,9). Classify the compatible reports and their repetition period.

Eight / twelve: classification	Tests / return period

Build a general solver

Can both schedules be right?

Advanced: gcd, divisibility and Bézout

Let $m,n>0$ and $d=\gcd(m,n)$. Supplied fact: if $\gcd(M,N)=1$, there are integers u,v with $uM+vN=1$. Write the reports as $t\equiv a \pmod m$, $t\equiv b \pmod n$.

5. Prove that a solution exists exactly when d divides $b-a$. When it exists, build one using the supplied fact. Prove uniqueness modulo mn/d , and explain how to choose the first nonnegative time.

Necessity

Construction and uniqueness

6. Use the construction for reports 7 modulo 10 and 1 modulo 6. Then create your own compatible pair with a shared factor and show your construction agrees with a direct check.

Given reports	Your compatible reports

Choose counts to compare

Can the next match hide between our answers?

Entry: rows and square arrays; algebra follows

A triangle has rows of lengths $1, 2, \dots, m$. A square has n rows of n counters. Both m and n are positive integers; matching totals satisfy $m(m+1)/2 = n^2$.

1. Find a match larger than one using at most 36 counters. Explain the triangle count with a picture or pairing argument. Why is adding one to both side labels not a reliable way to find the next match?

Triangle: rearrange or draw counters

Square: rearrange or draw counters

2. Set $x=2m+1$ and $y=2n$. Find an equation relating x and y . Prove both directions: a triangle-square match gives such a pair, and every positive integer pair satisfying your equation gives a match.

Match $\rightarrow$ equation

Equation $\rightarrow$ match

Test a machine that makes matches

Can the next match hide between our answers?

Algebra core: identities and growth

Proposed forward machine: $U(x,y)=(3x+4y,2x+3y)$. Work only with positive integer solutions of $x^2-2y^2=1$. A successful forward rule alone does not show that no solutions were skipped.

3. Prove that U preserves the equation and produces a strictly larger positive pair. Start with $(3,2)$, use the machine to reach a triangular-square total above one million, and check that total exactly.

x, y	m, n	Matching total	Check

4. Solve $X=3x+4y$ and $Y=2x+3y$ for x,y . Use that reverse machine on two generated examples. What must still be proved before it can certify that every possible match comes from the starting pair?

Reverse machine

Examples and missing proof

Make “next” a theorem

Can the next match hide between our answers?

Advanced proof: inequalities and integer descent

Use $D(x,y)=(3x-4y,3y-2x)$. Positive solutions have x odd and y even. Examine $(3,2)$ separately before treating $y>2$.

5. Prove that every positive solution other than $(3,2)$ has a positive predecessor with smaller y . Explain why repeated descent reaches $(3,2)$, and why reversing the steps proves that U lists every positive solution exactly once.

Inequalities and positivity

Descent, base case and completeness

6. Now decide whether your displayed total above one million is the first. Give the triangle row count and square side of the first match after it, and separate the roles of computation and proof.

First total above the threshold	Next match / exact certificate

Keep the cards, change multiplication

Choose a rule that never loses a symbol

Core: bits and distribution

Cards are $0=(0,0)$, $1=(1,0)$, $t=(0,1)$, $u=(1,1)=1+t$. Add coefficient bits modulo 2, so equal terms cancel. Multiply by distributing, then use ONE chosen rule: A has $t^2=t$; B has $t^2=t+1$. These are formal symbols, not residues 0,1,2,3 modulo 4.

1. Choose A or B and derive enough products to decide whether every nonzero card has a partner whose product is 1. Trade your evidence with a pair using the other rule.

0

0
0

1

1
0

t

0
1

u

1
1

A: $t^2=t$

×	0	1	t	u
0				
1				
t				
u				

B: $t^2=t+1$

×	0	1	t	u
0				
1				
t				
u				

2. For each rule, try to find two different inputs that multiplication by t sends to the same output. Explain either your collision or why no collision can exist.

Rule A

input $\times t$ output



Rule B

input $\times t$ output



Certify the arithmetic

Choose a rule that never loses a symbol

Algebra proof: quotient rings and inverses

Supplied fact: polynomials over the two-element field, reduced modulo a monic quadratic, form a commutative ring with identity; each class has a unique remainder $a+bt$. Associativity and distributivity come from polynomial arithmetic. A field is such a ring in which every nonzero element is invertible.

3. Identify the quadratic modulus for each rule. Explain why the facts supplied here together with your inverse work certify B as a field and reject A. Test the two possible roots of each modulus as another explanation.

Rule A: modulus and roots

Rule B: modulus and roots

4. Could a clever renaming make B into ordinary arithmetic modulo 4 while preserving both addition and multiplication? Give a property that any such renaming would have to preserve and show the mismatch.

Reasoning / examples / checks

Decode a hidden machine

Choose a rule that never loses a symbol

Continuation: affine equations over B

Use rule B from now on. A hidden machine has form $f(z)=az+b$, with a,b among $0,1,t,u$. Choose two distinct input cards and ask for their output cards. Arithmetic is the four-symbol arithmetic throughout.

5. Make an affine machine and give a partner two distinct input-output reports. Recover your partner's a,b . Prove that every pair of output reports at distinct inputs determines exactly one affine machine, including equal outputs.

Chosen input / output reports	Recover a and b; justify uniqueness

6. Decode the reports $f(1)=t$ and $f(t)=0$, then predict $f(u)$. Repeat these same reports under rule A: can they determine a unique machine? Use an explicit calculation.

Decode with B	Test the same reports with A

Certify the equation of a length

Which cube volumes can these tools reach?

Geometric entry: exact unit constructions

Start from a unit segment. Legal tools are an unmarked straightedge and compass, used for finitely many line/circle intersections. A physical measurement may suggest a length, but its construction needs an exact argument.

1. Choose and carry out an exact construction of $\sqrt{2}$ and then $\sqrt{3}$. Explain each length. If your $\sqrt{3}$ segment were the edge of a cube, what would its volume be? Does it solve the cube-volume 3 problem?

Exact construction and its certificate



unit = 1

2. A partner proposes the rational edge $1442/1000$ for volume 3. Check its cube exactly. Find a nearby rational edge whose cube lies on the other side of 3. Explain what the two checks establish and what they do not.

Exact cubes	Bounds and what they certify

What can one legal step add?

Which cube volumes can these tools reach?

Advanced: algebraic degree and supplied theorem

Supplied theorem: coordinates made from a unit segment by finitely many legal intersections lie in a tower of extensions of degree 1 or 2 over $\mathbb{Q}$. Hence any constructible real algebraic number has degree 2^r over $\mathbb{Q}$ for some integer $r \geq 0$. Degree means the degree of its irreducible minimal polynomial. This is a necessary condition, not a general sufficiency test.

3. Explain the quadratic step behind the theorem. If known coordinates lie in a field K , why do line-line, line-circle and two distinct circle intersections introduce at most a square root? Include tangency and intersections that do not exist.

Line-line / line-circle

Two circles / degenerate cases

4. Use the theorem to decide whether the exact positive edge α with $\alpha^3=3$ is constructible. Prove the needed irreducibility rather than merely calling α irrational.

Reasoning / examples / checks

Classify all integer requests

Which cube volumes can these tools reach?

Proof continuation: positive integer cube volumes

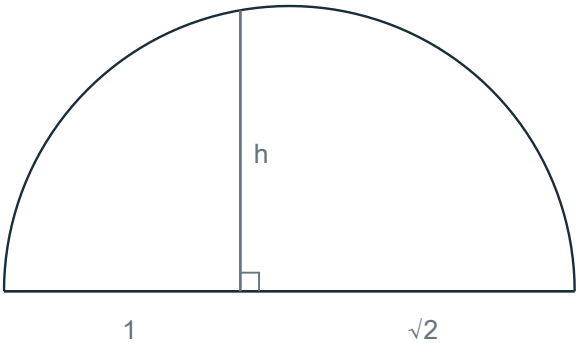
A request is a positive integer volume N , measured relative to the unit cube. Its positive edge is the real solution of $x^3=N$. Do not include negative volumes; if you discuss $N=0$, label it separately.

5. Classify all positive integers N for which the exact edge is constructible. Prove both directions. Test your classification on $N=8,9,27$ and 64 .

Classification / both directions	Check the supplied requests

6. A learner says, “Every irrational root is forbidden,” and another says, “The supplied theorem alone proves that every algebraic number of power-of-two degree is allowed.” Explain the flaw in each claim. Give an explicit constructible root of X^4-2 and certify it by legal square-root steps.

Use this diagram, or give your own construction



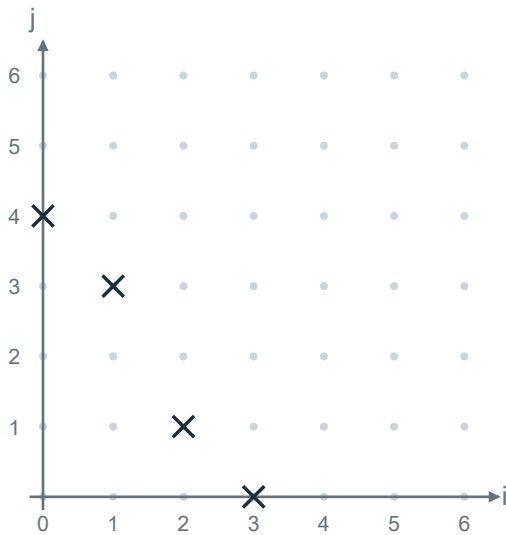
Find the places that survive

Move a corner, change the whole staircase

Core: grid directions and a complete boundary

Coordinates (i,j) are nonnegative integers. A corner (a,b) forbids itself and every (i,j) with $i \geq a$ and $j \geq b$. Corners are $(3,0), (2,1), (1,3), (0,4)$. The grid continues forever to the right and upward.

1. Choose a way to find every surviving position. Show your result on the grid and explain why enlarging the paper cannot reveal more survivors.



Boundary argument / witnesses

2. Is any of the four starting corners redundant, meaning that removing it leaves the forbidden region unchanged? Give a witness for each decision rather than judging the picture.

Reasoning / examples / checks

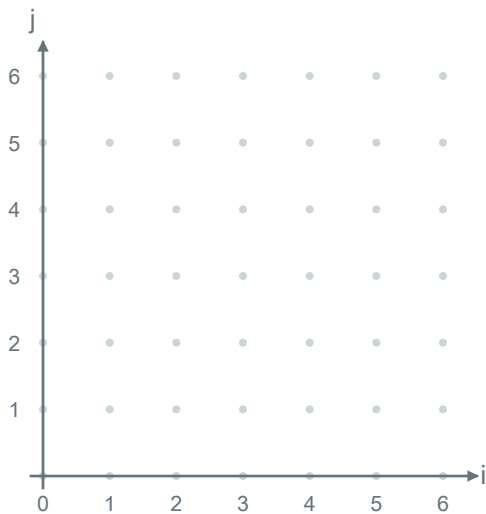
Save exactly two places

Move a corner, change the whole staircase

Core continuation: classify legal changes

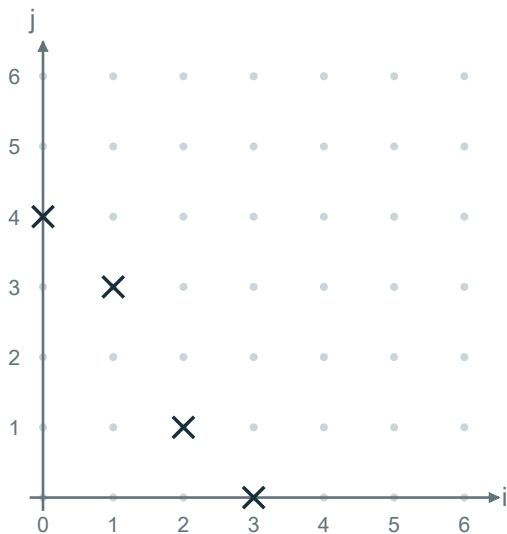
Move exactly one corner one unit right or one unit up. All other corners stay fixed. The goal is to keep only finitely many survivors and increase their number by exactly two. A moved corner may become redundant.

3. Find every move that meets the goal. Explain what happens for all eight possible moves, including moves that create infinitely many survivors.



Chosen changes and their effects

4. Return to the original corners. A right step or an up step sends a survivor to its neighboring position, or to zero if that neighbor is forbidden. Find every surviving position for which BOTH kinds of step go to zero. Explain why your list is complete.



Boundary argument / witnesses

What the staircase proves in algebra

Move a corner, change the whole staircase

Advanced: polynomial quotients and a basis argument

Work over $\mathbb{Q}$. Position (i,j) represents $x^i y^j$. Declare x^3, x^2y, xy^3, y^4 and all their polynomial multiples equal to zero. Thus we work in $\mathbb{Q}[x,y]/I$, where $I=(x^3, x^2y, xy^3, y^4)$. Do not erase terms merely because their total degree is large.

5. Prove that every polynomial has exactly one remainder using the surviving monomials. Give both spanning and independence arguments. Reduce $(x+y)^3$, and give an example showing that a simple total-degree cutoff is the wrong rule.

Spanning and independence

Reduction and cutoff counterexample

6. Classify every quotient element killed by multiplication by BOTH x and y . Could a linear combination of other survivors create a cancellation that produces an extra answer?

Reasoning / examples / checks

Make an unfamiliar arithmetic

Can one calculation remember a rate?

Algebra core: exact coefficient rules

A dual number is the formal pair $a+b\varepsilon$, with $a,b\in\mathbb{Q}$. Add coefficients separately. Multiply by distribution and replace ε^2 by 0. The pair $0+1\varepsilon$ is different from $0+0\varepsilon$. This is exact arithmetic in $\mathbb{Q}[T]/(T^2)$, not arithmetic with a tiny real ε .

1. Choose two nonzero dual numbers whose product is zero, and two whose product is 1. Then find an inverse for $3-2\varepsilon$ and explain which choices made your two searches possible.

Chosen factors	Ordinary coefficient	ε coefficient

2. Classify all invertible $a+b\varepsilon$. Explain exactly why canceling a nonzero factor can fail, even though canceling an invertible factor works.

Unit classification

Cancellation and a counterexample

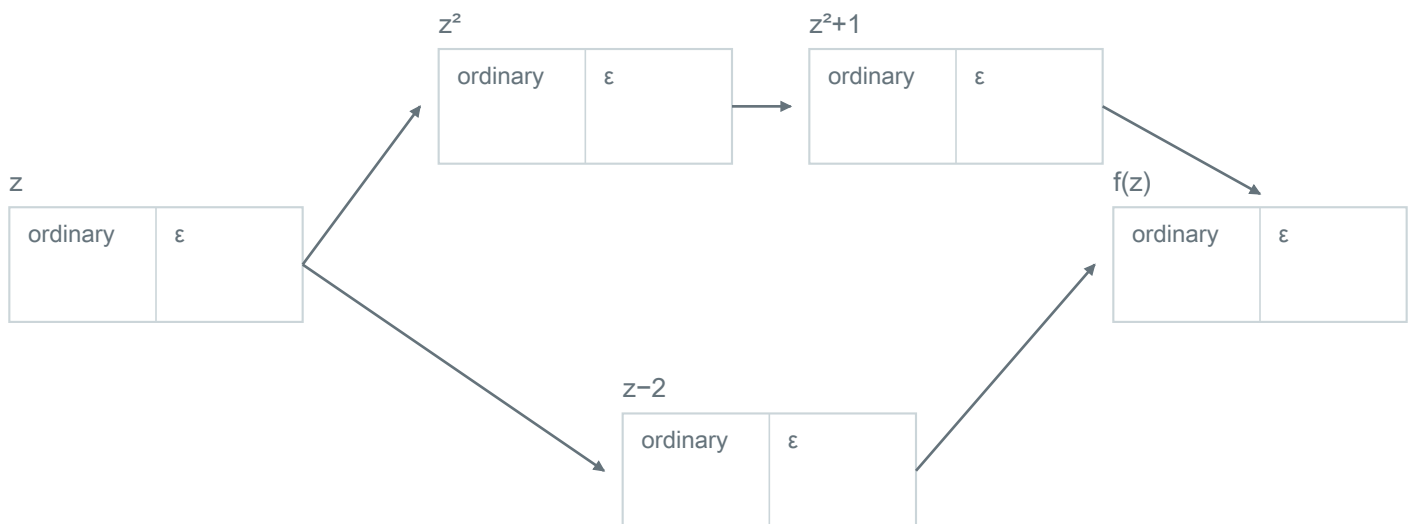
Run two numbers through one program

Can one calculation remember a rate?

Continuation: polynomial evaluation and derivatives

Use the polynomial program $f(z)=(z^2+1)(z-2)$. Evaluate it by its displayed operations, keeping both coefficients at every intermediate step. Polynomial derivative notation will be used on the next page.

3. Choose $z=3+\epsilon$ or $z=3+2\epsilon$ and run the program. Compare the ordinary output with $f(3)$. Predict how the ϵ coefficient changes if the input is $3+b\epsilon$ for any rational b .



4. A partner says the two output coefficients determine the whole polynomial. Choose two different polynomials that give the same output at $2+\epsilon$, and show an ordinary input at which they disagree. What information did the program forget?

Two polynomials / common dual output

An ordinary input that distinguishes them

Prove what the second coefficient means

Can one calculation remember a rate?

Algebra/calculus proof: polynomial power rule

For a polynomial $f(X) = \sum c_k X^k$ over $\mathbb{Q}$, define $f'(X) = \sum k c_k X^{k-1}$, with the constant term contributing 0. This agrees with the usual derivative for real polynomials. Prove statements for every polynomial, not just the chosen program. Supplied algebra fact: a nonzero polynomial over $\mathbb{Q}$ has only finitely many roots.

5. Prove $f(a+b\epsilon) = f(a) + b f'(a)\epsilon$ for every rational a, b . Use it to explain why the earlier program output 16 in its ϵ column when the input was $3+\epsilon$.

Reasoning / examples / checks

6. Use dual multiplication to derive the product rule for two polynomials f, g . Then derive the chain rule by evaluating $f(g(a+b\epsilon))$. State what would need to change to retain second-order information.

Product rule

Chain rule / second order

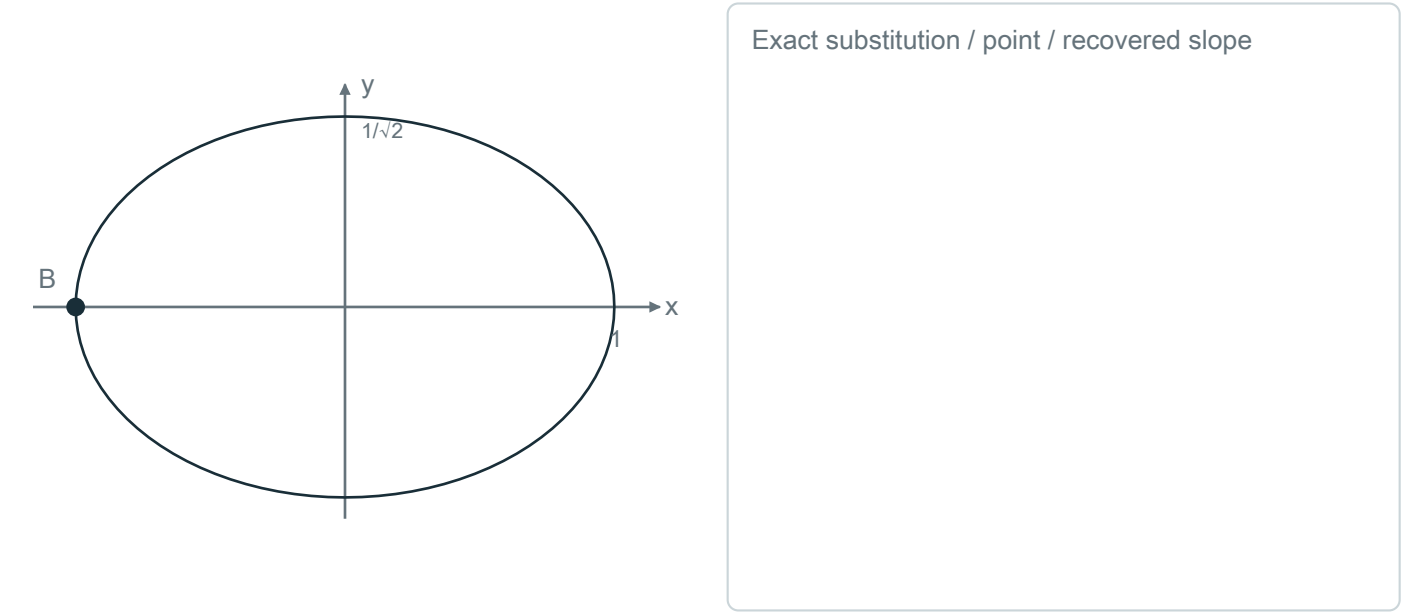
Aim, calculate and send

Encode a rational point with one slope

Core: slopes and exact fractions

The curve is $x^2+2y^2=1$. The marked point is $B=(-1,0)$. A nonvertical line of slope t through B has equation $y=t(x+1)$. Use exact fractions; a plotted approximation alone does not certify membership.

1. Choose $t=1$ or $t=2/3$, predict where the second intersection lies, and find it exactly by substitution. Give the point to a partner without the slope; let them try to recover your choice.



Exact substitution / point / recovered slope

2. Invent a different rational slope and create a point challenge. Then decode the supplied point $(-7/11,6/11)$, checking both its curve equation and the recovered line.

Your challenge	Decode the supplied point

Can any rational point escape?

Encode a rational point with one slope

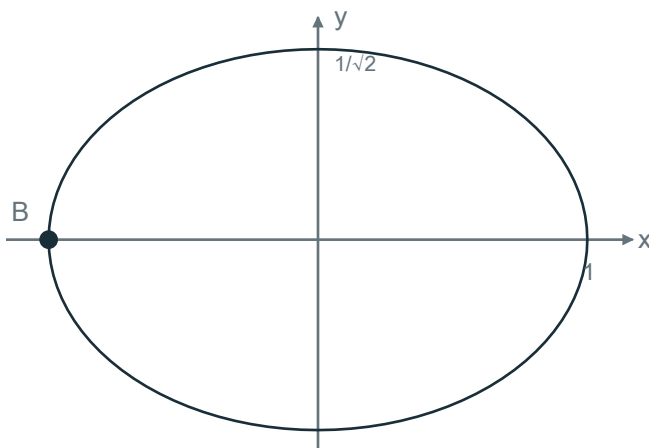
Core proof: known root and inverse

Continue to use finite rational slopes t . A complete parametrization claim needs both directions and every exceptional case.

3. Derive the second-intersection formula for general t . Prove that its denominator cannot vanish for rational t and that it always gives a point different from B . Does $t=0$ cause any difficulty?

Reasoning / examples / checks

4. Prove that every rational point on the ellipse other than B is recovered by exactly one rational slope. Describe the missing point geometrically and explain how an infinite-slope card completes the correspondence.



Exact substitution / point / recovered slope

Trade slopes for integer triples

Encode a rational point with one slope

Continuation: integer algebra and completeness

Write $t=p/q$ with integers p,q and $q\neq 0$. A point with rational coordinates can be written $(A/C,B/C)$, $C\neq 0$; scaling A,B,C together does not change the point.

5. Clear denominators in your parametrization to construct integer triples satisfying $A^2+2B^2=C^2$. Choose p,q to make a nontrivial example. Explain in what sense this produces every rational point, including any exception.

p, q and a triple	Completeness / exceptional point

6. Does a line through a known point guarantee a rational second point on every polynomial curve? Test $y^2=x^3+x+1$ through $(0,1)$ using the horizontal line $y=1$. Explain exactly which earlier step breaks.

Reasoning / examples / checks

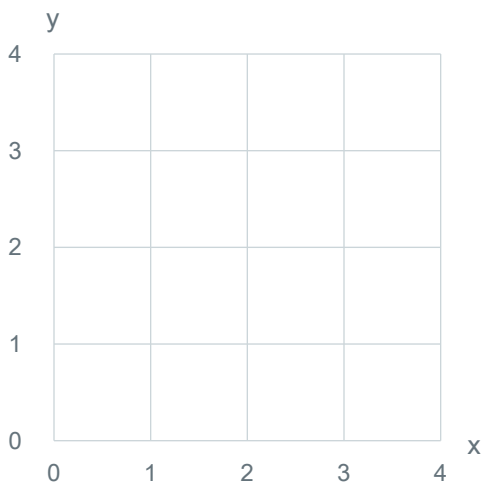
Find the finite point set

Can these finite points make one cycle?

Entry: residue arithmetic and an exhaustive search

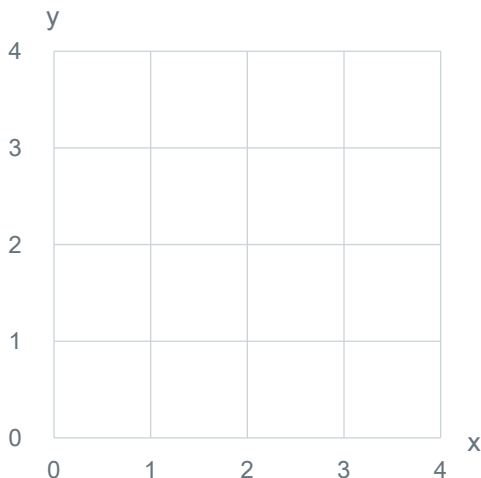
All arithmetic is modulo 5. Coordinates are residues 0,1,2,3,4. Place a point exactly when $y^2 = x^3 + x + 1$. The printed grid is a map of labels, not a sampled real curve: do not join the dots. Squares of 0,1,2,3,4 are 0,1,4,4,1.

1. Choose a search organized by columns or square values. Find every point, and give a certificate that none were missed. Predict which points must come in pairs before completing the search.



Column calculations / completeness certificate

2. Choose one coefficient change: replace the constant 1 by 0, or replace the coefficient of x by 0. Predict a difference, then find the new affine point count and explain why a smooth real-looking sketch would not settle it.



Column calculations / completeness certificate

Account for the point off the grid

Can these finite points make one cycle?

Advanced: projective coordinates, formal partial derivatives and a supplied theorem

Return to $y^2=x^3+x+1$. A projective point $[X:Y:Z]$ is a nonzero triple modulo nonzero scalar multiplication. The homogeneous equation is $Y^2Z=X^3+XZ^2+Z^3$. Supplied theorem: over a field of characteristic other than 2 or 3, a curve $y^2=x^3+ax+b$ with $\Delta=-16(4a^3+27b^2)\neq 0$, completed by $O=[0:1:0]$, has a commutative associative point-addition law with identity O . The next page supplies its formulas.

3. Prove that points with $Z\neq 0$ correspond exactly to the affine points already found, and find every projective point with $Z=0$. Check the discriminant hypothesis for the original curve.

$Z \neq 0$: normalize coordinates

$Z = 0$: solve, then identify scalings

4. Why should the theorem's nonzero-discriminant condition not be dropped? Test the curve $y^2=x^3$ at $(0,0)$ using its two partial derivatives. The proposed tangent slope for $y^2=x^3+ax+b$ is $s=(3x^2+a)/(2y)$. Identify what fails in this formula at the tested point.

Reasoning / examples / checks

Try repeated point addition

Can these finite points make one cycle?

Advanced: field division and the supplied group law

Use the original curve and arithmetic modulo 5. $O+P=P$. The inverse of (x,y) is $(x,-y)$; inverse pairs add to O . Otherwise, for $P=(x_1,y_1), Q=(x_2,y_2)$, use $s=(y_2-y_1)/(x_2-x_1)$ when $x_1 \neq x_2$; for $P=Q$ with $y_1 \neq 0$ use $s=(3x_1^2+1)/(2y_1)$. Then $P+Q=(x_3,y_3)$, where $x_3=s^2-x_1-x_2$ and $y_3=s(x_1-x_3)-y_1$. Doubling a point with $y=0$ gives O . Division means multiplying by a nonzero residue's inverse.

5. Choose a nonidentity point and repeatedly add it, starting at O . Record the orbit until it returns. Try to find a point that visits every point. If your first choice does not, choose another. Give at least one exact addition or doubling check.

Chosen point	Repeated sums / checks

6. For a point P whose orbit visits everything, prove that labeling its multiples by integers modulo the return time turns the point law into modular addition. Explain which part is experimental evidence and which part uses the supplied theorem.

Reasoning / examples / checks

Solve equations in the hidden clock

Can these finite points make one cycle?

Advanced continuation: a cyclic-group representation

Use the generator $P=(0,1)$ and the orbit you proved. A point equation may now be translated into an equation on its multiple-of- P label. A solution list must include every point or explain why none can work.

7. Solve $2Q=P$, then solve $3Q=P$. Solve $3Q=O$ and $3Q=(2,1)$. Translate every answer back to actual point coordinates and explain the difference among the four equations.

$$2Q=P$$

labels $\rightarrow$ points; why the list is complete

$$3Q=P$$

labels $\rightarrow$ points; why the list is complete

$$3Q=O$$

labels $\rightarrow$ points; why the list is complete

$$3Q=(2,1)$$

labels $\rightarrow$ points; why the list is complete

Choose who moves farther

Design the round when closeness begins

Core: fractions and simultaneous movement

At each integer round, $x'=(1-\alpha)x+\alpha y$ and $y'=\beta x+(1-\beta)y$. Both right sides use the OLD pair. Start at $(-3,9)$. Choose rule A: $\alpha=1/4,\beta=1/2$; or rule B: $\alpha=1/2,\beta=1/4$.

1. Run your chosen rule for two rounds. Compare with a partner using the other rule. Predict the long-term location and explain why averaging the two current positions may mislead.



Rule / round	x	y	What seems unchanged?

2. Find a nonzero weighted sum $px+qy$ that never changes under rule A. Then find one for rule B. Prove your choices work for every starting pair, and explain why equal shrinkage of the gap does not force equal limits.

Rule A: weighted certificate

Rule B: weighted certificate

Predict a first round, then design one

Design the round when closeness begins

Algebra core: a formula and sharp inequalities

Use rule A. Write $d=x-y$ and $L=(2x+y)/3$. “Within ϵ ” means BOTH $|x-L|\leq\epsilon$ and $|y-L|\leq\epsilon$. Round 0 is the initial pair.

3. Derive x_n and y_n from L and d_0 without repeated updating. For the start $(-3,9)$, find the FIRST round when both coordinates are within $1/100$ of L . Prove the preceding round fails. Can they be exactly equal at a finite round?

Exact formula	First round / preceding round

4. Design an initial pair whose limit is 2 and whose FIRST round within $1/8$ is round 3. Find one pair, then classify every real initial pair with this property.

Chosen pair	All pairs / boundary cases

Choose the dynamics

Design the round when closeness begins

Continuation: real parameters and geometric limits

Now allow $0 \leq \alpha, \beta \leq 1$. Initial coordinates may be any real numbers. A common limit means both coordinates converge to the same finite number.

5. For unequal initial coordinates, classify every parameter pair that produces a common limit. Include zero movement, swapping and exact finite meeting. Derive the limit when it exists.

Parameter case	Limit / proof

6. Return to the start $(-3,9)$. Design ALL parameter pairs that give common limit 3 and halve the absolute gap each round. Which preserve the order of the markers and which alternate it?

Parameter equations	Order of the markers

Slide one image arrow

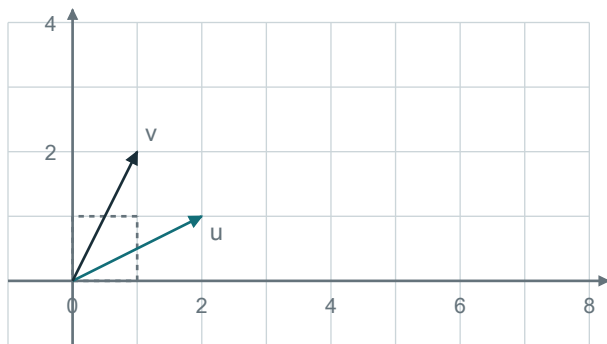
Can you preserve area without preserving shape?

Entry: vectors, orientation and parallelograms

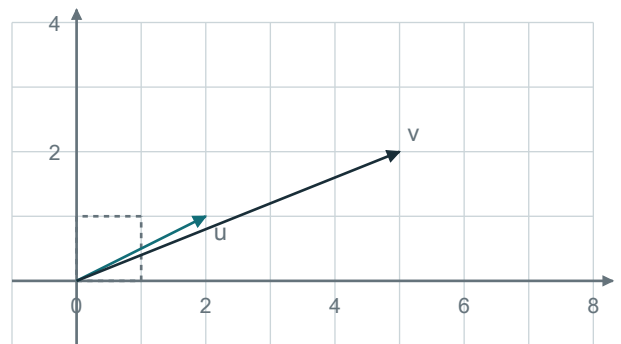
The unit horizontal arrow maps to $u=(2,1)$; the unit vertical arrow maps to $v=(k,2)$. The full linear rule is $A_k(x,y)=x u+y v$. Traverse the image square in the order $0,u,u+v,v$. Counterclockwise traversal has positive signed area, clockwise negative; a collapsed image has signed area 0.

1. Choose $k=1$ or $k=5$. Draw the image square, find its ordinary area, and determine its signed area. Compare your result with the other choice. Justify area by geometry, not by counting partly covered grid squares.

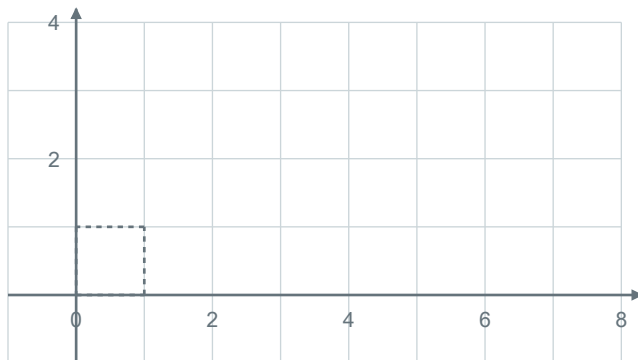
k = 1



k = 5



2. Test $k=4$. Find two distinct inputs with the same output. Describe the whole image and ALL inputs producing an arbitrary reachable output. Explain why there cannot be an inverse on the plane.



Collision / all preimages

Turn the picture into a test

Can you preserve area without preserving shape?

Proof gate: dot products and invertibility

For vectors $p=(p_1,p_2), q=(q_1,q_2)$, the dot product is $p \cdot q = p_1q_1 + p_2q_2$. Supplied geometry fact: projection onto a unit normal gives signed perpendicular height. The left normal to a nonzero vector (a,c) is $(-c,a)/\sqrt{a^2+c^2}$.

3. For any real matrix $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, prove that the signed area of its ordered image square is $ad-bc$ and its ordinary area is $|ad-bc|$. Include the case $(a,c)=(0,0)$. Explain exactly when the image collapses.

Nonzero first column

Zero column / collapse criterion

4. Find ALL real k for which A_k preserves ordinary area of parallelograms. For each, give an explicit inverse and its orientation. Prove that every other k except the collapse value is still invertible.

All area-preserving choices	Inverse and orientation

Compose, then test rigidity

Can you preserve area without preserving shape?

Continuation: matrix products and a length identity

Write $\det M=ad-bc$ for $M=\begin{bmatrix}a&b\\c&d\end{bmatrix}$. A shear $S_t(x,y)=(x+ty,y)$ has matrix $\begin{bmatrix}1&t\\0&1\end{bmatrix}$. In $A_3 S_t$, the shear acts first.

5. Choose a nonzero t and compare $A_3 S_t$ with $S_t A_3$ on the two coordinate arrows. Prove for arbitrary real 2×2 matrices M,N that $\det(MN)=\det M\cdot\det N$, then explain why both compositions preserve area even when they differ.

Coordinate-arrow images	Product certificate

6. Area preservation alone does not preserve every length: demonstrate that with A_3 . Now decide whether a linear map can preserve ordinary area AND the lengths of both unit coordinate arrows, yet fail to preserve some other length. Prove your answer.

A₃ length test	Both unit lengths: proof

Try to hide and recover directions

Choose bases that reveal what an arrow does

Advanced core: exact linear algebra

Over $\mathbb{Q}$, let $T:\mathbb{Q}^4\rightarrow\mathbb{Q}^3$ be $T(x_1,x_2,x_3,x_4)=(x_1+x_2,x_3+x_4,x_1+x_2+x_3+x_4)$. You may choose a basis independently in the source and target. A basis must be proved independent and spanning.

1. Invent two distinct inputs with the same output. Decide whether $(3,-2,1)$ and $(3,-2,0)$ are reachable. Then describe ALL inputs giving any reachable target.

Chosen inputs / targets	All preimages

2. Choose source and target bases so the matrix has as few nonzero entries as possible. Show each basis vector's image, prove both lists are bases, and explain why fewer nonzero entries cannot suffice.

Chosen source vector	Its image under T	Target-basis coordinates

Prove the inventory does not depend on your choices

Choose bases that reveal what an arrow does

General proof: basis extension over a field

A block $k \rightarrow 0$ is one killed source direction; $k \rightarrow k$ with identity is one transmitted direction; $0 \rightarrow k$ is one unreachable target direction. A direct sum places blocks side by side in both source and target. A nonzero representation is indecomposable if it is not a direct sum of two nonzero representations.

3. For ANY linear map $V \rightarrow W$ between finite-dimensional spaces over a field k , construct bases giving only these blocks. Prove your chosen transmitted source vectors really are independent modulo the kernel, and account for every direction.

$$k \longrightarrow 0 \qquad k \longrightarrow k \qquad 0 \longrightarrow k$$

Construction / independence / spanning

4. Why are the three displayed block types indecomposable? Prove that any two decompositions have the same number of each type. Do equal inventories also guarantee equivalence of two one-arrow representations?

Indecomposable blocks

Uniqueness / converse

Design an arrow and test the rule’s boundary

Choose bases that reveal what an arrow does

Continuation: classification versus similarity

Keep source Q^4 and target Q^3 . Later in this page, a SAME-basis change for an endomorphism means replacing a square matrix M by $P^{-1}MP$, with the same invertible P on both sides.

5. Design a map $Q^4 \rightarrow Q^3$ with exactly one transmitted direction. Give a basis certificate. Then list every possible killed/transmitted/unreachable inventory for these fixed dimensions and construct a map for each.

Chosen source vector	Its image under T	Target-basis coordinates

6. Someone applies your theorem to say I_2 and $2I_2$ become equal under a SAME-basis change. Audit this claim. Show how independent source/target bases can identify the arrows, and why a same-basis change cannot.

Independent bases

Same basis on both sides

Choose a partner for one matrix

Find what commutes, then explain the cancellations

Advanced core: products on column vectors

Work over $\mathbb{Q}$. Let $N = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$ and $C = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$. For any matrices A, B define $[A, B] = AB - BA$. Matrix multiplication acts right to left: ABv means apply B first, then A .

1. Choose a nonscalar matrix B that commutes with N , and another that does not. Certify the first by full products and expose the second with an explicit input vector. Check N^2 as a useful clue.

NB

v

→

B

→

N

→

BN

v

→

N

→

B

→

Products / chosen matrices / vector test

2. Classify EVERY $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ satisfying $[N, B] = 0$. Express your answer using I and N , and explain why a few successful trials would not prove completeness.

Products / equations	Complete commuting family

Audit the new operation

Find what commutes, then explain the cancellations

Proof gate: distinguish two operations

Keep ordinary multiplication and the bracket separate. The trace of a square matrix is the sum of its diagonal entries. A Lie algebra bracket is bilinear, has $[A,A]=0$, and obeys the Jacobi identity studied on the next page.

3. Prove bilinearity and $[A,A]=0$ for arbitrary rational 2×2 matrices. Prove also that every commutator has trace 0. Must a zero bracket mean either matrix is zero?

Bilinearity / alternation

Trace / zero bracket

4. Compare $[[N,C],C]$ with $[N,[C,C]]$. Use the result to decide whether the bracket is associative. Explain why this does not contradict associativity of ordinary matrix multiplication.

$[[N,C],C]$

$[N,[C,C]]$

Why every Jacobi sum vanishes

Find what commutes, then explain the cancellations

General proof: ordered triple products

For arbitrary matrices A, B, D of the same size, the Jacobi expression is $[[A, B], D] + [[B, D], A] + [[D, A], B]$. Ordinary products remain in their original order; associativity lets us omit parentheses inside each three-letter product.

5. Prove the Jacobi identity for ALL rational square matrices by expanding the expression and pairing cancellations. Then prove the related identity $[A, BD] = [A, B]D + B[A, D]$. State where associativity of multiplication is used.

$[[A, B], D]$

$[[B, D], A]$

$[[D, A], B]$

6. Call a matrix central if it commutes with EVERY rational 2×2 matrix. Prove that the central matrices are exactly scalar multiples of I . Use only the tests with N and C for necessity, then verify sufficiency. What is the center inside the trace-zero matrices?

Necessary tests / sufficiency

Inside trace-zero matrices

Build and improve the catalog

Make a catalog that remembers every choice

Core: named choices and color agreement

People: Aster and Birch are red; Cedar and Dahlia are blue; Elm is green. Badges: r,s are red; t is blue; u,v are green; w is gold. A legal entry is an ordered (person,badge) pair with matching colors. Names remain distinct even when colors agree.

1. Choose a catalog layout and record EVERY legal entry, once each. Prove completeness. Identify any person or badge that appears in no entry. Can one slot per color preserve both names?

People	Aster red	Birch red	Cedar blue	Dahlia blue	Elm green	
Badges	r red	s red	t blue	u green	v green	w gold

Your catalog: choose a layout; keep both names

2. You may recolor ONLY badge w to red, blue or green. Choose a color to make the catalog as large as possible. Find every best choice and certify that no permitted recoloring can do better.

New color of w	Effect / maximality

Send tokens without losing either record

Make a catalog that remembers every choice

Continuation: functions and forced placements

Return w to gold. A token can carry one person name and one badge name. Tokens may repeat the same assignments, and several tokens may land in the same catalog entry. A placement must preserve both names.

3. Create compatible assignments for at least three tokens, including two tokens with identical records. Place them in your catalog and explain why the placements are forced. Then create an incompatible token assignment and explain exactly why no placement exists.

Token name	Person record	Badge record	Catalog destination

4. For arbitrary sets A, B, C and functions $f: A \rightarrow C, g: B \rightarrow C$, define $P = \{(a, b) : f(a) = g(b)\}$. Its projections return the first and second records. Given any set X and functions $u: X \rightarrow A, v: X \rightarrow B$ with $f \circ u = g \circ v$, prove there is EXACTLY ONE $h: X \rightarrow P$ with projections u, v . Include X empty.

Existence: define the map

Uniqueness / empty set

Test a rival catalog

Make a catalog that remembers every choice

Proof continuation: uniqueness of the object

A rival catalog Q has two record maps $q_A: Q \rightarrow A, q_B: Q \rightarrow B$ whose colors agree. It has the universal property if EVERY compatible token assignment has exactly one record-preserving map into Q . Merely containing a record for every compatible pair is not enough.

5. Start with your catalog P . Make Q by adding a second, distinct copy of $(Aster, r)$, with identical person and badge records. Which part of the universal property fails? Give the smallest token test. Also explain what fails if the original $(Aster, r)$ entry is removed instead.

Duplicate an entry

Remove an entry

6. Suppose a rival Q really does have the universal property. Prove that there is exactly one bijection $P \rightarrow Q$ preserving both records. Prove it is a bijection using the two universal properties, without comparing cardinalities.

Maps in both directions

Composites / uniqueness

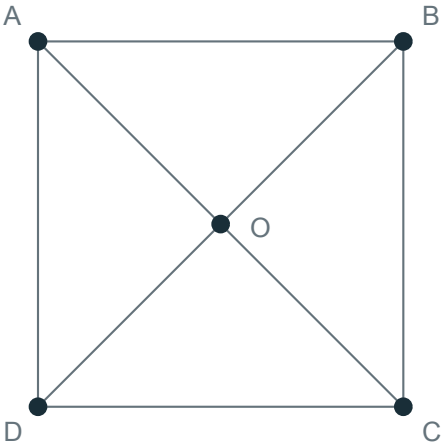
Make a closed pattern

Which face cards erase the same edge pattern?

Core: parity on an exact graph

Vertices A,B,C,D form a square in that order; O is its center. Edges are AB,BC,CD,DA,OA,OB,OC,OD. O is a vertex, not an overpass. A pattern marks any subset of edges and is CLOSED when each vertex meets an even number of marked edges. Toggling means marked↔unmarked. Initially the only available face moves toggle the three edges of OAB or OCD.

1. Choose a nonempty closed pattern other than a single triangle boundary. Check every vertex. Try each available face move and explain why any legal sequence keeps the pattern closed.



Available face cards (not selected edges)

OAB

OCD

Chosen marks / parity / move record

2. Find EVERY closed pattern without checking all subsets separately. Let a,b,c,d be the marks on AB,BC,CD,DA. What must the four spoke marks be? Prove every choice of a,b,c,d gives exactly one closed pattern.

Outer marks a,b,c,d	Spokes / completeness proof

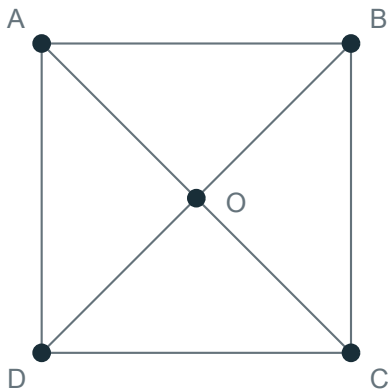
Classify what the available cards can change

Which face cards erase the same edge pattern?

Core proof: an invariant and a construction

Keep only face cards OAB and OCD available. Two closed patterns count as equivalent when some legal sequence turns one into the other. A card may be used repeatedly; twice has no net effect.

3. Classify ALL equivalence classes of closed patterns. Give both a certificate preventing moves between different classes and a procedure connecting any two patterns in the same class. Do not stop after finding one loop that cannot be erased.



Invariant and a move sequence

4. Choose a target to erase: the outer square, or the closed loop $AB+BC+OC+OA$. You may buy missing face cards OBC and ODA. Find ALL minimum purchases for each target, with a legal sequence and a reason fewer cards cannot work.

Outer square: purchases

$AB+BC+OC+OA$: purchases

Name the quotient you have computed

Which face cards erase the same edge pattern?

Advanced: F2 boundaries and homology

Over F_2 , C_0 has the five vertices as basis, C_1 the eight edges, and C_2 only the available filled triangles OAB, OCD . The boundary ∂_1 of an edge is the sum of its endpoints; ∂_2 of a triangle is the sum of its boundary edges. Define $H_1 = \ker \partial_1 / \text{im } \partial_2$ once you have proved $\text{im } \partial_2 \subseteq \ker \partial_1$. Use vertex order A, B, C, D, O ; edge order $AB, BC, CD, DA, OA, OB, OC, OD$; and face order OAB, OCD .

5. Write the two boundary matrices in the displayed basis orders and verify $\partial_1 \partial_2 = 0$. Use your earlier classification to give bases and dimensions for $\ker \partial_1$ and $\text{im } \partial_2$. Identify H_1 explicitly and say what its zero element means in the game.

∂_1

ABBC CDDA OAOB OC OD

A

B

C

D

O

∂_2

OAB OCD

AB

BC

CD

DA

OA

OB

OC

OD

Bases / product check

6. Now choose ANY subset S of the four possible triangular face cards. Classify the resulting quotient without recomputing a large matrix each time. Which cycle coordinates survive, and when can every closed pattern be erased?

Chosen face set S	Surviving coordinates / proof

Let two controls distinguish equal totals

How much free space completes this module?

Advanced core: actual module action

Let $R = Q \times Q$ with componentwise ring operations and identity $(1, 1)$. On $M(a, b) = Q^a \oplus Q^b$, let $(r, s) \cdot (u, v) = (ru, sv)$. Here a, b are nonnegative integers. An R -linear isomorphism is an invertible additive map preserving this action. Set $e_1 = (1, 0), e_2 = (0, 1)$.

1. Choose two different dimension pairs with the same total dimension and test whether their modules can be R -isomorphic. Then compare $P = M(3, 1)$ and $Q = M(1, 3)$. Find a control-based invariant that proves your answer, rather than relying on the names of the two components.

$P = M(3, 1)$



r control



s control

Circles mark basis directions.

Test e_1 and e_2 ; explain the invariant.

$Q = M(1, 3)$



r control



s control

Circles mark basis directions.

Test e_1 and e_2 ; explain the invariant.

2. For ANY finitely generated unital R -module M , prove $M = e_1 M \oplus e_2 M$ and show both summands are finite-dimensional rational vector spaces. Prove that the ordered pair of dimensions completely classifies M , including the converse construction of an R -isomorphism.

Idempotent decomposition

Finite dimensions / classification

Buy the smallest free complement

How much free space completes this module?

Advanced: free versus projective modules

A finite free R -module is R^n , $n \geq 0$. In this investigation “finitely generated projective” means a module that has a complement C with $M \oplus C \cong R^n$ for some finite n . Direct sum adds the two component dimensions.

3. Find the smallest n for which $P=M(3,1)$ has a complement in R^n , and exhibit that complement. Then solve this problem for EVERY $M(a,b)$. Use the result to prove every finitely generated R -module is projective and classify which are free.

Module $M(a,b)$	Complement you choose	Free result / minimality

4. Can adding the SAME module to $P=M(3,1)$ and $Q=M(1,3)$ make them isomorphic? Can adding some free R^n make $M(1,0)$ free? Prove both answers, and then find a nonfree addition that does make $M(1,0)$ free.

The same addition to P and Q

Free / nonfree additions

Allow differences without inventing negative spaces

How much free space completes this module?

Advanced: the universal group completion

The monoid of projective isomorphism classes has direct sum as addition. Its K_0 is the abelian group generated by symbols $[M]$, subject to $[M \oplus N] = [M] + [N]$ and $[0] = 0$. An actual class $[M]$ and a formal difference $[P] - [Q]$ are different kinds of description.

5. Compute $z = [M(3,1)] - [M(1,3)]$. Invent a different pair of actual modules giving the same z and justify equality using the relations. Decide whether z is the class of any actual module. More generally, identify exactly which signed pairs can be represented by an actual module.

Equivalent formal descriptions	Which are actual classes?

6. Prove $K_0(R) \cong \mathbb{Z}^2$, rather than merely giving a map to $\mathbb{Z}^2$. Show every signed pair occurs and that coordinate differences determine formal classes uniquely. Then prove the universal statement: every additive map $\mathbb{N}^2 \rightarrow G$ to an abelian group extends uniquely to a group homomorphism $\mathbb{Z}^2 \rightarrow G$.

Isomorphism to $\mathbb{Z}^2$	Universal extension / uniqueness

Build without counting the same turn twice

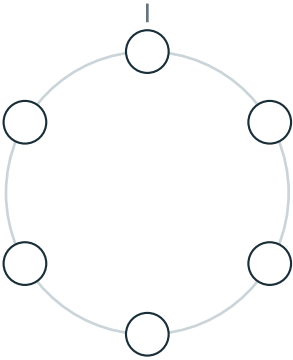
Count necklaces when every bead count is fixed

Core: six beads and a precise equivalence rule

Use exactly three red beads and three blue beads on six positions. Two necklaces are the same if a rotation makes all named colors match. Do not flip the ring or exchange color names. For recording, 1 means red and 0 means blue; strings read clockwise from a temporary top mark.

1. Choose a catalog method and find every different necklace. Prove that none is missing and that no two entries are rotations of one another. A bare list without a completeness argument is not enough.

temporary top mark



record clockwise

Open catalog / why complete?

2. Restore the top mark. Count all labeled strings with three red beads. For each necklace in your catalog, count its DISTINCT rotations. Explain why dividing the labeled count by 6 does not give the necklace count.

Your necklace	Distinct turns / counting check

Count the patterns fixed by each turn

Count necklaces when every bead count is fixed

Proof core: count the same pairs twice

Let r_j move each bead forward j positions modulo 6, for $j=0,\dots,5$. A labeled pattern is FIXED by r_j when the resulting labeled string is unchanged. Count only strings with exactly three red beads.

3. Find the number of patterns fixed by every r_j , with a reason for each turn type. Average the six counts and compare with your catalog. Why must a fixed pattern have constant color along each position-cycle of the turn?

j	Position cycles	Fixed patterns / reason
0		
1		
2		
3		
4		
5		

4. Explain why this average MUST count necklaces. Count pairs (turn, labeled pattern fixed by that turn) in two ways. Prove that the patterns in any one rotation orbit contribute exactly 6 pairs, even when the orbit has fewer than 6 members.

Count by turns

Count by orbits

Change the rules, then generalize

Count necklaces when every bead count is fixed

Continuation: reflections and gcd

For the first task on this page ONLY, allow turning the ring over as well as rotating it, while keeping color names fixed. For the next task return to rotations only.

5. Which of your six-bead necklaces merge when reflections become legal? Give an actual reflected pair and prove the new complete count. Explain why a coincidence for four beads would not settle this case.

An actual reflected pair	Complete revised catalog

6. For $n \geq 1$ positions and exactly r red beads, $0 \leq r \leq n$, derive the number fixed by a j -position turn using $g = \gcd(n, j)$. State a rotations-only counting formula, justify it, and use it for eight positions with four red beads.

General fixed-pattern formula	Eight positions / four red